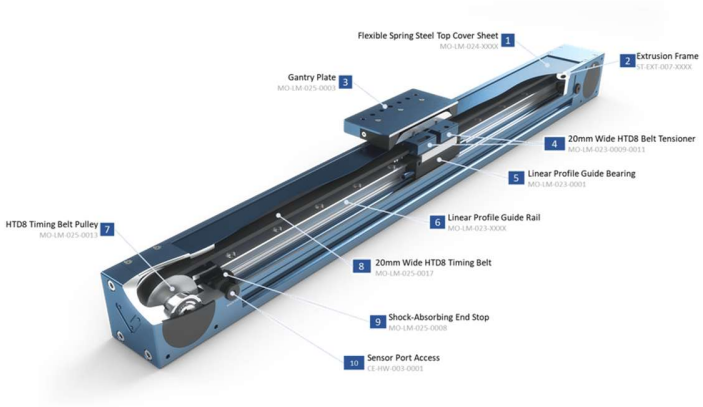
Actuator Sizing

Axis 1(X-Axis) holds: Second belt assembly, Power-screw assembly, external load.

$$m_{total} \approx 2.07\,Kg$$



$$\sum T = J\alpha$$

$$\therefore T_m - T_L = (J_m + J_L)\,\alpha$$

Inertia of load

$$J_L = mr^2$$

Where:

m = Total mass of Load

r = radius of pulley

Using a standard pulley: r = 0.637 cm

$$\therefore J_L = 2070 \times 0.673^2 = 840\,g \cdot cm^2$$

For unknown motor, we assume $J_m = J_l$:

$$J_{tot} = 2 \times 840 = 1680\,g \cdot cm^2$$

Resistive Torque

To calculate the resistive torque, we use:

$$\boldsymbol{T_L = mg\mu r}$$

Where:

m = Total mass of load

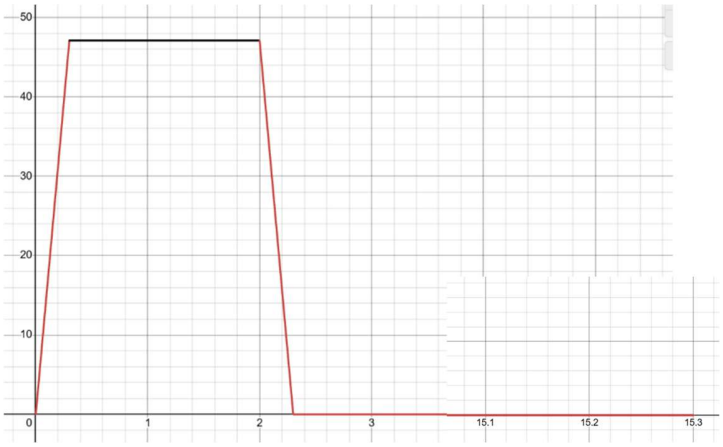
g = Gravitational acceleration

μ = Coefficient of friction

r = radius of pulley

$$\therefore \boldsymbol{T_L = 2.07 \times 9.81 \times 0.05 \times 0.637 = 0.65\, N \cdot cm}$$

Velocity Profile



1. **From** t = 0.0s **to** t = 0.30s:

   Constant acceleration: $\alpha = 157\ \text{rad} \cdot \text{s}^{-2}$

2. **From** t = 0.30s **to** t = 2.0s:

   Constant speed: $\omega = 47.1\ \text{rad} \cdot \text{s}^{-1}$

3. **From** t = 2.0s **to** t = 2.3s:

   Constant acceleration: $\alpha = -157\ \text{rad} \cdot \text{s}^{-2}$

4. **From** t = 2.3s **to** t = 15.3s:

   Constant speed: $\omega = 0\ \text{rad} \cdot \text{s}^{-1}$

Torque

$$\boldsymbol{T_m = (0.65 + 1680 \times 10^{-5}\,\alpha)\ N \cdot cm}$$

$$T_1 = 0.65 + 1680 \times 10^{-5} \times 157 = \boldsymbol{3.2\ N \cdot cm}$$

$$T_2 = 0.65 + 1680 \times 10^{-5} \times 0 = \boldsymbol{0.65\ N \cdot cm}$$

$$T_3 = 0.65 + 1680 \times 10^{-5} \times -157 = \boldsymbol{-2.0\ N \cdot cm}$$

$$T_4 = 0.65 + 1680 \times 10^{-5} \times 0 = \boldsymbol{0.65 N \cdot cm}$$

$$\boldsymbol{T_{r.m.s} = \sqrt{\frac{T_1^2 \times t_1 + T_2^2 \times t_2 + T_3^2 \times t_3 + T_4^2 \times t_4}{t_1 + t_2 + t_3 + t_4}}}$$

$$\therefore T_{r.m.s} = \sqrt{\frac{3.2^2 \times 0.30 + 0.65^2 \times 1.70 + (-2.0)^2 \times 0.30 + 0.65^2 \times 13}{15.3}} = \boldsymbol{1.28\ N \cdot cm}$$

Note: t_4 is the time it takes the Z-axis to complete two cycles (grip and pick up the product)

Choice

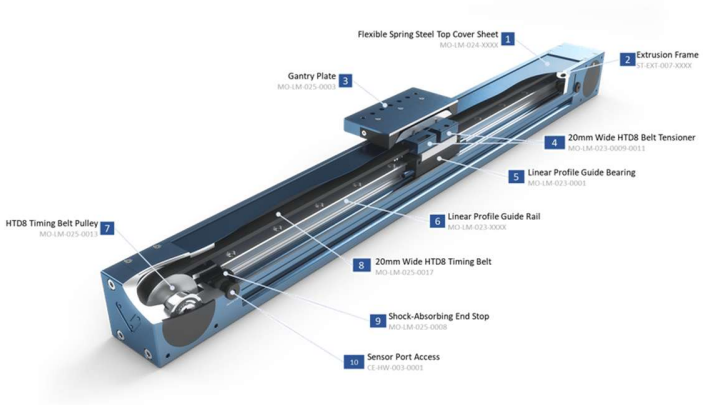
We decided to pick NEMA17 stepper motor as our actuator for the following reasons:

1. Both the r.m.s. and maximum torques calculated above fall within the operating limits of the actuator.
2. Our assumed motor inertia is much larger than NEMA17's rotor inertia, meaning there is no doubt that the NEMA17 can easily control the axis.
3. Stepper motors provide open-loop control over position, which is needed for controlling the position of the robot.



Axis 2 (Y-Axis) holds: Power-screw assembly, external load.

$$m_{total} \approx 1.0\,Kg$$



$$\sum T = J\alpha$$

$$\therefore T_m - T_L = (J_m + J_L)\,\alpha$$

Inertia of load

$$J_L = mr^2$$

Where:

m = Total mass of Load

r = radius of pulley

Using a standard pulley: r = 0.637 cm

$$\therefore J_L = 1000 \times 0.673^2 = 405.7\,g \cdot cm^2$$

For unknown motor, we assume $J_m = J_l$:

$$J_{tot} = 2 \times 405.7 = 811.4\,g \cdot cm^2$$

Resistive Torque

To calculate the resistive torque, we use:

$$\boldsymbol{T_L = mg\mu r}$$

Where:

m = Total mass of load

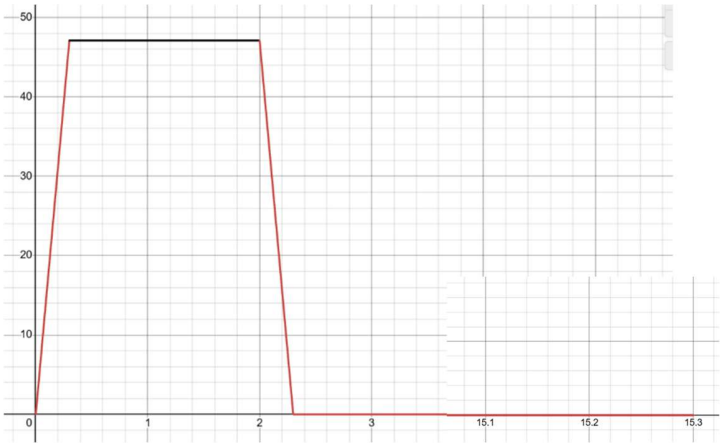
g = Gravitational acceleration

μ = Coefficient of friction

r = radius of pulley

$$\therefore \boldsymbol{T_L = 1.0 \times 9.81 \times 0.05 \times 0.637 = 0.31\, N \cdot cm}$$

Velocity Profile



1. **From** t = 0.0s **to** t = 0.30s:

   Constant acceleration: $\alpha = 157\ \text{rad} \cdot \text{s}^{-2}$

2. **From** t = 0.30s **to** t = 2.0s:

   Constant speed: $\omega = 47.1\ \text{rad} \cdot \text{s}^{-1}$

3. **From** t = 2.0s **to** t = 2.3s:

   Constant acceleration: $\alpha = -157\ \text{rad} \cdot \text{s}^{-2}$

4. **From** t = 2.3s **to** t = 15.3s:

   Constant speed: $\omega = 0\ \text{rad} \cdot \text{s}^{-1}$

Torque

$$\boldsymbol{T_m = (0.65 + 811.4 \times 10^{-5}\,\alpha)\ N \cdot cm}$$

$$T_1 = 0.65 + 811.4 \times 10^{-5} \times 157 = \boldsymbol{1.6\ N \cdot cm}$$

$$T_2 = 0.65 + 811.4 \times 10^{-5} \times 0 = \boldsymbol{0.65\ N \cdot cm}$$

$$T_3 = 0.65 + 811.4 \times 10^{-5} \times -157 = \boldsymbol{-0.97\ N \cdot cm}$$

$$T_4 = 0.65 + 811.4 \times 10^{-5} \times 0 = \boldsymbol{0.65 N \cdot cm}$$

$$\boldsymbol{T_{r.m.s} = \sqrt{\frac{T_1^2 \times t_1 + T_2^2 \times t_2 + T_3^2 \times t_3 + T_4^2 \times t_4}{t_1 + t_2 + t_3 + t_4}}}$$

$$\therefore T_{r.m.s} = \sqrt{\frac{1.6^2 \times 0.30 + 0.65^2 \times 1.70 + (-0.97)^2 \times 0.30 + 0.65^2 \times 13}{15.3}} = \boldsymbol{0.62\ N \cdot cm}$$

Note: t_4 is the time it takes the Z-axis to complete two cycles (grip and pick up the product)

Choice

We decided to pick NEMA17 stepper motor as our actuator for the reasons stated before.

Axis 3 (Z-Axis) holds: Vacuum pump, external load.

$$m_{total} \approx 0.4\,Kg$$



$$\sum T = J\alpha$$

$$\therefore T_m - T_L = (J_m + J_L)\,\alpha$$

Inertia of load

$$J_L = J_{screw} + J_{load} = (\frac{1}{2} \times m_{screw} \times r_{screw}^2) + (m_{load} \times (\frac{P}{2\pi})^2)$$

Where:

P = Pitch of screw

Using a standard 20 cm screw:

$$\therefore J_L = 6.4 \times 10^{-7} + 0.4 \times (\frac{0.08}{2\pi})^2 = 12.9\,N \cdot cm^2$$

For unknown motor, we assume $J_m = J_l$:

$$J_{tot} = 2 \times 12.9 = 25.8\,g \cdot cm^2$$

Resistive Torque

To calculate the resistive torque, we use:

$$T_L = \frac{mgP}{2\pi\eta}$$

Where:

m = Total mass of load

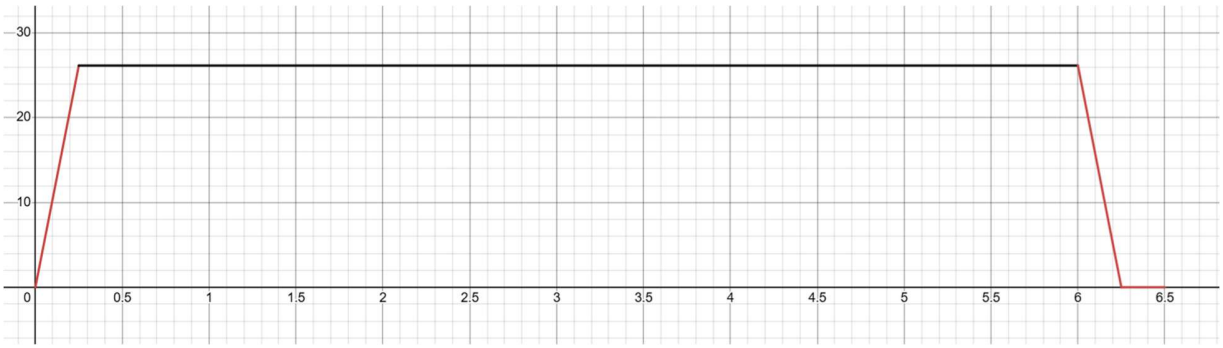
g = Gravitational acceleration

P = Pitch of Screw

η = Efficiency

$$\therefore T_L = \frac{0.4 \times 9.81 \times 0.008}{2\pi \times 0.5} = 1\ N \cdot cm$$

Velocity Profile



1. **From** t = 0.0s **to** t = 0.25s:

   Constant acceleration: $\alpha = 104.7\ \text{rad} \cdot \text{s}^{-2}$

2. **From** t = 0.25s **to** t = 6.0s:

   Constant speed: $\omega = 26.18\ \text{rad} \cdot \text{s}^{-1}$

3. **From** t = 6.0s **to** t = 6.25s:

   Constant acceleration: $\alpha = -104.7\ \text{rad} \cdot \text{s}^{-2}$

4. **From** t = 6.25s **to** t = 6.5s:

   Constant speed: $\omega = 0\ \text{rad} \cdot \text{s}^{-1}$

Torque

$$\boldsymbol{T_m = (1 + 25.8 \times 10^{-5}\,\alpha)\ N \cdot cm}$$

$$T_1 = 1 + 25.8 \times 10^{-5} \times 104.7 = \boldsymbol{1.027\ N \cdot cm} \qquad T_3 = 1 + 25.8 \times 10^{-5} \times -104.7 = \boldsymbol{0.973\ N \cdot cm}$$

$$T_2 = 1 + 25.8 \times 10^{-5} \times 0 = \boldsymbol{1.0\ N \cdot cm} \qquad T_4 = 1 + 25.8 \times 10^{-5} \times 0 = \boldsymbol{1.0\ N \cdot cm}$$

$$\boldsymbol{T_{r.m.s} = \sqrt{\frac{T_1^2 \times t_1 + T_2^2 \times t_2 + T_3^2 \times t_3 + T_4^2 \times t_4}{t_1 + t_2 + t_3 + t_4}}}$$

$$\therefore T_{r.m.s} = \sqrt{\frac{1.027^2 \times 0.25 + 1.0^2 \times 5.75 + 0.973^2 \times 0.25 + 1^2 \times 0.25}{15.3}} = \boldsymbol{1.0\ N \cdot cm}$$

Choice

We decided to pick **25GA-370 motor with encoder** as our actuator for the following reasons:

1. Both the r.m.s. and maximum torques calculated above fall within the operating limits of the actuator.
2. Our assumed motor inertia is larger than rotor inertia, meaning there is no doubt that the actuator can easily control the axis.
3. This motor will allow us to perform PID control on the Z-axis.



End Effector

Our Choice of an end effector is a vacuum pump with a suction cup diameter of 25mm.

$$\boldsymbol{F = P \cdot A}$$

Where:

P = Pressure difference provided by pump

A = Area of suction cup

Required force is equivalent to the weight of the product; Assuming the mass of the product is 50g, we get:

$$0.050 \times 9.81 = \mathrm{P} \times \frac{\pi}{4} \times (25 \times 10^{-3})^2$$

Solving this we get that the required pressure difference P = 1.0 Kpa.

Choice

A 12V vacuum pump able to deliver 66Kpa of negative pressure, which makes it more that suitable for the job!



Feeding

The feeder is a rack and pinion mechanism that should be able to push 8 products at once.

As we already know:

$$T_m = T_L + (J_m + J_L)\,\alpha$$

Inertia of load

Modeling the rack as a cuboid with dimentions $1 \times 2 \times 15$ cm

And the pinion as a cylinder with diameter = 10 cm and thickness = 1 cm

And switching the load from translational to effective rotational inertia we get:

$$J_L = \frac{1}{2} \times (700 \times \pi \times 0.05^2 \times 0.01) \times 0.05^2 + (700 \times 0.15 \times 0.02 \times 0.01) \times 0.05^2 + 8 \times 0.05 \times 0.05^2 = 1.12 \times 10^{-3}\ Kg \cdot m^2$$

For unknown motor, we assume $J_m = J_l$:

$$J_{tot} = 2 \times 1.12 \times 10^{-3} = 2.24 \times 10^{-3}\ g \cdot cm^2$$

Resistive Torque

To calculate the resistive torque, we use:

$$T_L = mg\mu r$$

Where:

m = Total mass of load

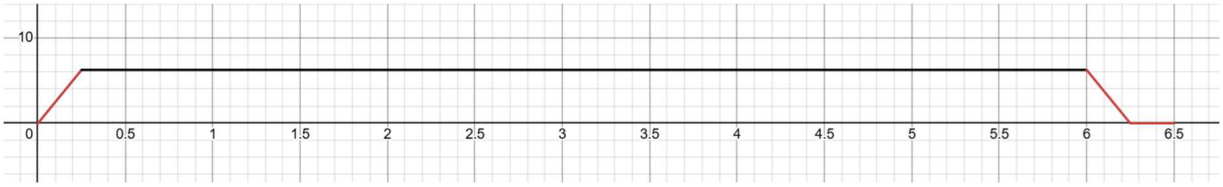
g = Gravitational acceleration

μ = Coefficient of friction

r = radius of pinion

$$\therefore T_L = 8 \times 0.05 \times 9.81 \times 0.5 \times 0.05 = 0.0981\ N \cdot m$$

Velocity Profile



1. **From** t = 0.0s **to** t = 0.25s:

   Constant acceleration: $\alpha = 25\ \text{rad} \cdot \text{s}^{-2}$

2. **From** t = 0.25s **to** t = 6.0s:

   Constant speed: $\omega = 6.16\ \text{rad} \cdot \text{s}^{-1}$

3. **From** t = 6.0s **to** t = 6.25s:

   Constant acceleration: $\alpha = -25\ \text{rad} \cdot \text{s}^{-2}$

4. **From** t = 6.25s **to** t = 6.5s:

   Constant speed: $\omega = 0\ \text{rad} \cdot \text{s}^{-1}$

Torque

$$\boldsymbol{T_m = (0.0981 + 2.24 \times 10^{-3}\,\alpha)\ N \cdot m}$$

$$T_1 = 0.0981 + 2.24 \times 10^{-3} \times 25 = \boldsymbol{0.1541\ N \cdot m}$$

$$T_2 = 0.0981 + 2.24 \times 10^{-3} \times 0 = \boldsymbol{0.0981\ N \cdot m}$$

$$T_3 = 0.0981 + 2.24 \times 10^{-3} \times 25 = \boldsymbol{0.1541\ N \cdot m}$$

$$T_4 = 0.0981 + 2.24 \times 10^{-3} \times 0 = \boldsymbol{0.0421\ N \cdot m}$$

$$\boldsymbol{T_{r.m.s} = \sqrt{\frac{T_1^2 \times t_1 + T_2^2 \times t_2 + T_3^2 \times t_3 + T_4^2 \times t_4}{t_1 + t_2 + t_3 + t_4}}}$$

$$\therefore T_{r.m.s} = \sqrt{\frac{0.1541^2 \times 0.25 + 0.0981^2 \times 5.75 + 0.1541^2 \times 0.25 + 0.0421^2 \times 0.25}{15.3}}$$

$$= \boldsymbol{0.0981\ N \cdot m = 1.027\ Kgf \cdot cm}$$

Choice

We decided to pick **MG996 Servo** as our actuator for the following reasons:



1. Both the r.m.s. and maximum torques calculated above fall within the operating limits of the actuator.
2. Our assumed motor inertia is larger than rotor inertia, meaning there is no doubt that the actuator can easily control the axis.
3. This motor will allow us to control the distance moved by our pusher in order to feed the parts one at a time.